

Aarushi Kotwani

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SUMMARY

Data Professional with 2+ years of hands-on experience in **Python, SQL, AWS, and modern BI tools**, specializing in **data pipelines, feature engineering, and predictive modeling**. Experienced in designing **end-to-end analytical systems**, optimizing performance, and deploying scalable, production-ready solutions. Passionate about building robust architectures that enable high-quality, data-driven insights.

EXPERIENCE

Data Scientist (Capstone)

Jan 2025 – Jun 2025

Amazon

Seattle, WA, USA

- Built an end-to-end **data engineering + ML pipeline** in Python and AWS (S3, Glue, Lambda, Redshift) to automate ingestion, validation, feature creation, and forecasting for 500+ Climate Pledge signatories, improving data processing efficiency by **35%**.
- Developed and optimized **Scikit-learn regression and time-series models** to analyze emission trends, engineered domain-specific features, and identified high-impact sustainability drivers through statistical analysis.
- Created technical **Tableau dashboards** integrating model outputs to visualize carbon reduction trajectories, detect anomalies, and forecast progress toward 2040 net-zero targets.

Data Analyst Intern

Jun 2024 – Jun 2025

Seattle University

Seattle, WA, USA

- Conducted exploratory analysis and regression modeling on multi-year institutional spend data, revealing **\$80K+** cost-saving opportunities.
- Automated Power BI dashboards and SQL-based ETL pipelines, reducing manual reporting time by **15%**.
- Delivered forecasting models improving annual budget accuracy by **15%**, empowering leadership to make data-backed allocation decisions.

Data Analyst

Apr 2022 – Aug 2023

Freelancer

India

- Analyzed customer and operational datasets for **3–5 clients**, performing data cleaning, transformation, and exploratory analysis using **Python (pandas, NumPy), SQL, and Excel** to support business reporting needs.
- Developed **lightweight text-analysis workflows** using NLTK, TF-IDF, and keyword extraction to classify customer comments, identify common issues, and measure sentiment trends.
- Built **dashboards and automated reports** in Tableau/Power BI that visualized key metrics, uncovered patterns, and streamlined clients' decision-making processes.

SKILLS

- Languages:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras), R, SQL
- Machine Learning & AI:** Regression, Classification, Clustering, Neural Networks (CNN, LSTM), NLP, XGBoost, Forecasting (ARIMA, Prophet), LLMs, LangChain
- Tools:** AWS (S3, Glue, Redshift, Lambda), Tableau, Power BI, Git, Docker,
- Data Engineering:** ETL Pipelines, Hadoop, Spark, Data Cleaning, Feature Engineering, Model Deployment.
- Analytics:** A/B Testing, Hypothesis Testing, Statistical Modeling, Data Visualization, ArcGIS
- Core Strengths:** Problem Solving, Data Storytelling, Cross-functional Collaboration, Strategic Insight

PROJECTS

RAG-Based Document Chatbot

Nov 2025 – Dec 2025

- Built a RAG-based chatbot that answers queries strictly from uploaded PDFs by implementing text extraction, chunking, embeddings, and semantic retrieval using LangChain, OpenAI, and ChromaDB, with a Streamlit UI for real-time interaction.
- Achieved 100% context relevance by testing in-scope vs. out-of-scope queries and optimizing retrieval parameters to deliver accurate, hallucination-free responses.

Impact of Environmental and Health Factors on Mental Health

Oct 2024 – Dec 2024

- Constructed** a Bidirectional LSTM model (**$R^2 = 0.904$**) to forecast the prevalence of mental disorders using preprocessed global datasets (1990–2019).
- Enhanced** data quality for **100,000+ records** through MinMax scaling and outlier removal, improving feature reliability and model accuracy.

Bird Species Classification Using Neural Networks

Apr 2024 – Jun 2024

- Architected custom **Convolutional Neural Networks (CNNs)** to classify bird species from spectrogram data, achieving **100% accuracy** in binary classification and **99.21% accuracy** in multi-class classification.
- Processed audio data using Librosa and TensorFlow/Keras, optimizing for noise reduction and limited labeled data.

EDUCATION

Master of Science in Data Science, Seattle University

Sep 2023 – Jun 2025

Master of Arts in Economics, Bundelkhand University

Aug 2020 – Jun 2022

Master of Commerce, Bundelkhand University

Aug 2017 – Jun 2019